

# A Comparison of High-Power GaAs pHEMT Dies for Flip-Chip Interposer Applications

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**Abstract**—This paper presents a research-based comparison of three high-power GaAs pHEMT MMIC dies suitable for power amplifier applications exceeding 3 W in output power. In addition to electrical performance, the dies are compared for compatibility with a flip-chip design that employs at least 75  $\mu\text{m}$  gold bumps, aiming to facilitate low-cost manufacturing with an approximate target price of \$50 or less per unit in bulk. A concise recommendation is provided, based on power level, operating frequency, current requirements, ease of flip-chip integration, and expected unit cost.

## I. INTRODUCTION

Applications in wireless communication, radar, and satellite systems often demand power amplifiers that can deliver high output power (exceeding 3 W) at microwave or millimeter-wave frequencies. The choice of a GaAs pHEMT die is influenced by factors such as:

- Required output power and linearity
- Operating frequency and gain
- Package/die pad dimensions and assembly method
- Cost constraints and per-unit price

This work compares three commercially available GaAs pHEMT dies:

- **MACOM MAAP-011139-DIE** (GaAs pHEMT die)
- **Mini-Circuits PMA5-123-3W-D<sup>+</sup>** (GaAs MMIC die)
- **Qorvo TGA2533** (GaAs pHEMT die)

Each device is evaluated on output power, current handling, pad dimensions, feasibility for flip-chip ball bonding, and approximate pricing.

## II. DIE COMPARISONS

### A. Electrical Performance and Frequency Range

**MACOM MAAP-011139-DIE** operates from about 28.5 GHz up to 31 GHz, delivering up to approximately 36.5 dBm of saturated power (about 4.4 W). Typical biasing involves  $V_D \approx 6$  V with drain currents above 3 A under maximum drive, making it suitable for applications where high current and millimeter-wave frequencies are acceptable.

**Mini-Circuits PMA5-123-3W-D<sup>+</sup>** covers 8–12 GHz, producing about 35.5 dBm (3.5 W) of saturated power at around 1.25 A of current when biased at 7 V. Its comparatively high gain (near 30 dB) is attractive in lower Ku-band systems or other radar and communications setups operating below 12 GHz.

**Qorvo TGA2533** is specified for 12.7–15.4 GHz and provides around 34–35 dBm of output power (about 2.5–3.2 W). Typical biasing is at 6 V and about 1.3 A quiescent, rising under power drive. This device meets just above the 3 W benchmark at saturation, making it a borderline fit for designs strictly requiring  $> 3$  W continuous.

### B. Dimensions and Pad Size for Flip-Chip

All three dies have overall dimensions suitable for mounting on an interposer:

- **MAAP-011139-DIE**: Approximately  $3.1 \times 2.8$  mm, typical pHEMT wire-bond design. Bond pads are sufficiently large (on the order of 100  $\mu\text{m}$  or more) to be adapted to flip-chip if needed.
- **Mini-Circuits PMA5-123-3W-D<sup>+</sup>**: About  $3.4 \times 2.8$  mm, with bond-pad dimensions generally larger than 75  $\mu\text{m}$ , which is adequate for the gold bump flip-chip method. The device is primarily intended for wire bonding but can potentially be adapted.
- **Qorvo TGA2533**: Around  $3.2 \times 3.0$  mm, with typical bond pads on the order of 100  $\mu\text{m}$  width. The datasheet indicates a recommendation for flip-chip ball bonding.

For the first 2 dies the datasheets predominantly describe wire-bond assembly, each die's bond pads generally exceed 75  $\mu\text{m}$ , making flip-chip feasible for custom interposer designs if careful attention is paid to bump alignment, final die thickness, and gold bump standoff.

### C. Current Handling and Assembly Feasibility

All three devices can handle  $\geq 1$  A:

- **MAAP-011139-DIE** draws several amps at full power (potentially exceeding 3 A).
- **PMA5-123-3W-D<sup>+</sup>** typically draws about 1.25 A.
- **TGA2533** typically draws around 1.3 A quiescent, up to about 1.7 A under drive.

### D. Approximate Pricing

At sufficiently large purchase volumes, these devices can be priced near or below \$50 per unit:

- **MACOM MAAP-011139-DIE**: Noted to be \$25 at moderate or higher volumes.
- **Mini-Circuits PMA5-123-3W-D<sup>+</sup>** and **Qorvo TGA2533**: Achieve near or below \$50 at high-quantity purchases, according to supplier websites.

For the last 2 at lower volumes, the cost may exceed \$50, but bulk procurement typically brings unit costs near the target range.

### III. CONCLUSION AND RECOMMENDATION

For applications demanding over 3 W at microwave frequencies where a flip-chip interposer is desirable, these three GaAs pHEMT dies can each offer a viable solution:

- **MAAP-011139-DIE** provides the highest saturated power (over 4 W), making it the best choice when maximum power in the 28–31 GHz region is prioritized.
- **PMA5-123-3W-D<sup>+</sup>** offers a solid balance of 3.5 W output and relatively high gain from 8–12 GHz, drawing moderate current.
- **TGA2533** reaches just beyond 3 W near 15 GHz, with relatively modest current requirements around 1.3–1.7 A.

If minimizing cost per die ( $\sim$  \$50) and achieving at least 3 W across a broad portion of the Ku-band are top priorities, **Mini-Circuits PMA5-123-3W-D<sup>+</sup>** (for 8–12 GHz) or **Qorvo TGA2533** (for 12.7–15.4 GHz) stand out. However, for higher-frequency millimeter-wave applications above 28 GHz, **MACOM MAAP-011139-DIE** provides the best power headroom. All three can be integrated via flip-chip ball bonding using  $\sim$  75  $\mu$ m gold bumps, although wire bonding remains the most directly supported assembly. Ultimately, the choice depends on the targeted frequency band, required power margin, and system current budget.

### REFERENCES

- [1] MAAP-011139-DIE Data Sheet, MACOM Technology Solutions.
- [2] PMA5-123-3W-D<sup>+</sup> Data Sheet, Mini-Circuits.
- [3] TGA2533 Data Sheet, Qorvo.